Completeness of the Bethe Ansatz

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These notes are inspired by a talk of Jiang Yunfeng (https://www.koushare.com/live/details/11521).

1 Background

For an introduction to the Bethe ansatz see [2]; for the completeness problem of the Bethe ansatz that we care about, see [1].

In short, we are concerned with the solutions of the following equations (the Bethe equations):

$$\Big(\frac{\lambda_k + i/2}{\lambda_k - i/2}\Big)^L = \prod_{j\neq k} \frac{\lambda_k - \lambda_j + i}{\lambda_k - \lambda_j - i} \quad (k = 1, \dots, N),$$

together with some additional conditions:

- The equations are symmetric under permutation of the λ_i; we regard roots that differ by a permutation as the same root.
- The λ_i are pairwise distinct.
- If a root has $\lambda_1 = i/2$, then necessarily $\lambda_2 = -i/2$, and we require in addition that

$$\prod_{k=3,\dots,N} \Big(-\frac{\lambda_k + i/2}{\lambda_k - i/2}\Big)^L = 1.$$

Assume $N < L/2$. If the Bethe ansatz is complete, then the number of solutions equals $\binom{L}{N} - \binom{L}{N-1}$. Our idea is that, for a system of algebraic equations, if the multiplicity of a root can be defined correctly, then the number of roots is determined by certain highest-degree terms of the system, namely by the BKK theorem. The multiplicity is equal to the dimension of the Bethe vector. Consider the following example:

$$(d^2/dx^2 - u)f = 0.$$

Its characteristic equation is $\lambda^2 - t = 0$. When $t \neq 0$, $\lambda = \pm\sqrt{t}$, and each root corresponds to a one-dimensional space. When $t = 0$, the root $\lambda = 0$ has multiplicity 2, corresponding to a two-dimensional solution space.

As an example, we have the following guess: if $(\lambda_1, \dots, \lambda_N)$ is a solution of the Bethe equations with $\lambda_1 = \lambda_2$, and if this solution has multiplicity 3, then one can construct a Bethe vector from this root.

1.1 Roots at infinity

Given a system of algebraic equations, the "expected" number of solutions can be computed via the BKK theorem: if the coefficients of the system satisfy a suitable open condition, then the number of solutions equals the mixed volume.

We relax this condition as follows: if the zero loci $f_i = 0$ intersect transversally, one can add "roots at infinity" so that the BKK theorem holds. For equations in n unknowns, one compactifies $\mathbb{C}^n$ so that intersection theory applies: the Zariski closure of $f_i = 0$ defines a homology class, and the expected number of solutions equals the cup product of these classes. If a solution lies exactly in the part added by the compactification, we call it a root at infinity.

2 The case $N = 2$

Write

$$f(\lambda_1, \lambda_2) = (\lambda_1 + i/2)^L(\lambda_1 - \lambda_2 - i) - (\lambda_1 - i/2)^L(\lambda_1 - \lambda_2 + i).$$

Replace λ_1, λ_2 by x, y. The Bethe equations become

$$f(x, y) = 0, \qquad f(y, x) = 0.$$

We want to apply the BKK theorem. First compute the Newton polytope, i.e. expand f. For even L, the Newton polytope of $f(x, y)$ is the convex hull of $(0, 0), (0, L), (1, L-1), (1, 1)$; for odd L it is the convex hull of $(0, 0), (0, L), (1, L-1), (0, 1)$. The Newton polytope of $f(y, x)$ is the mirror image under $x = y$. A computation gives a mixed volume equal to L^2, so there are L^2 solutions in total.

Consider the number of solutions with $x = y$, i.e. the system $f(x, y) = 0$, $x = y$. It is easy to check that $f(x, y)$ is a polynomial of degree L, so there are L solutions. Hence the number of solutions with $x \neq y$ is $(L^2 - L)/2$.

Next we treat the case $x = i/2$, $y = -i/2$. We claim that this solution carries an extra multiplicity L, which is introduced when deriving the equations. The dimension of the genuine Bethe vector equals this multiplicity minus L (note that this guess differs from the preceding condition), so the number of solutions agrees with the number predicted by the earlier conjecture.

2.1 The multiplicity of the root $(i/2, -i/2)$

The multiplicity of a root is defined as $\dim_{\mathbb{C}}(\mathcal{O}_p/\mathcal{I})$; for the meaning of the notation see GTM 52. Make the translation $x \to x - i/2$, $y \to y + i/2$. The system becomes

$$f_1(x, y) = (x + i)^L(x - y) - x^L(x - y + 2i), f_2(x, y) = y^L(y - x - 2i) - (y - i)^L(y - x).$$

For the case we care about, the multiplicity equals the dimension of $\mathbb{C}[[x, y]]/\langle f_1, f_2\rangle$. The polynomial f_1 is linear in y; let y_0 be the root of $f(x, y_0) = 0$. A computation gives $y_0 \in \mathbb{C}[[x]]$, $y_0 = x - 2i^{-L+1}x^L + o(x^L)$. Make the change of variables $y_{\text{new}} = y - y_0$; then $f_1 = (x^L - (x+i)^L)y_{\text{new}}$. Note that in the local ring $x^L - (x + i)^L$ is invertible, so the ideal generated by f_1 is the ideal generated by y_{new}.

Then $\mathbb{C}[[x, y]]/\langle f_1, f_2\rangle = \mathbb{C}[[x]]/\langle f_2(x, y_0)\rangle$. First compute $f_2(x, y_0) \bmod x^{L+1} = -2i(1-(-1)^L)x^L$. Hence for odd L the multiplicity is L. We next compute the case of even L: $f_2(x, y_0) \bmod x^{L+2}$ (TBD), hence the multiplicity is $L+1$. This agrees with our prediction: the dimension of the Bethe vector at the root $(i/2, -i/2)$ differs from the multiplicity by L.

3 The case $N = 3$

In this section we explain in detail the difficulty that arises for $N = 3$. We use the following modified inclusion–exclusion:

Theorem 1. *Let $S \subset \mathbb{R}^3$ be a finite set (with coordinates x, y, z). Write $S_{x=y}$ for $\{(x, y, z) \in S \mid x = y\}$. Then the number of elements of S with x, y, z pairwise distinct is*

$$\#S - \#S_{x=y} - \#S_{y=z} - \#S_{x=z} + 2\#S_{x=y=z}.$$

For $N > 3$ see https://math.stackexchange.com/questions/3659385/.

Define

$$f(x, y, z) = \big((x + i/2)^L (x - y - i)(x - z - i) - (x - i/2)^L (x - y + i)(x - z + i)\big)/i.$$

Note that $f(x, y, z)$ is symmetric in y, z. The Bethe equations are $f(x, y, z) = f(y, z, x) = f(z, x, y) = 0$.

The highest-degree terms of f are $(L - 4)x^{L+1} - (L - 2)(y + z)x^L + Lyzx^{L-1}$. The vertices of its Newton polytope are

$$(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1), (1, 1, L - 1), (1, 0, L), (0, 1, L), (0, 0, L + 1).$$

Hence $f(x, y, z) = f(y, z, x) = f(z, x, y) = 0$ has $(L+1)^3$ solutions, $f(x, y, z) = f(z, x, y) = x-y = 0$ has $(L + 1)^2$ solutions, and $f(x, y, z) = x - z = x - y = 0$ has $L + 1$ solutions. Substituting into the formula above, there are $\binom{L+1}{3}$ solutions with x, y, z pairwise distinct.

3.1 The multiplicity of the root $(i/2, -i/2, z)$

Substituting $x = i/2$, $y = -i/2$ gives $f(x, y, z) = f(y, z, x) = 0$. Now $f(z, x, y) = (z - i/2)(z + i/2)(z^{L-1} + \cdots)$; apart from $z = \pm i/2$ there remain $L - 1$ roots. We guess that each such root has extra multiplicity L, so the number of solutions should be $\binom{L+1}{3} - L(L - 1) = \binom{L}{3} - \binom{L}{2}$, in agreement with the expectation.

Remark 2. Some assumptions were made to make the above argument work; we collect them here:

- The BKK theorem requires the coefficients of the equations to satisfy some open conditions. In fact the Bethe equations do not satisfy these conditions; for example, for $L = 6$, $N = 3$, assuming $x = y = z$ gives $f = 3x^5 - \frac{5}{2}x^3 + \frac{3}{16}x$, while the BKK theorem predicts 7 solutions. But we guess that the argument above still holds, because all roots at infinity lie on a hyperplane $\lambda_i = \lambda_j$, and thus do not affect the result.
- The solution $(-i/2, i/2, i/2)$ has the same multiplicity in the two systems $f(x, y, z) = f(y, z, x) = f(z, x, y) = 0$ and $f(x, y, z) = f(y, z, x) = z - y = 0$.

4 The case $N \geq 4$

For $N = 4$, suppose the solution is (x, y, z, w). By the BKK theorem the number of solutions is $\binom{L+2}{4}$. Based on some numerical results we guess that the number of solutions of the form

$(-i/2, i/2, z, w)$ with $z, w \neq \pm i/2$ is $\binom{L-1}{2}$, while $(-i/2, i/2, i/2, w)$ with $w \neq \pm i/2$ does not contribute to the multiplicity. Finally the multiplicity of $(-i/2, -i/2, i/2, i/2)$ is $\binom{L}{2}$.

The final number of solutions is

$$\binom{L+2}{4} - \binom{L-1}{2} \cdot L - \binom{L}{2} = \binom{L}{4} - \binom{L}{3},$$

in agreement with the expectation.

For general L, N, define $S(L,N) = \binom{L}{N} - \binom{L}{N-1}$. We expect that the number of solutions can be obtained as follows: suppose that $\lambda_1 = \cdots = \lambda_k = -i/2$, $\lambda_{k+1} = \cdots = \lambda_{2k} = -i/2$, and $\lambda_{2k+1}, \ldots, \lambda_N \neq \pm i/2$. Let the number of such solutions be $B(k; L, N)$, with multiplicity $M(k; L, N)$. Then, as expected,

$$S(L,N) = \binom{L+N-2}{N} - \sum_{k=1}^{L/2} B(k; L, N) \cdot M(k; L, N).$$

Based on the above results we may make the following guess: $B(k; L, N) = \binom{L-1}{N-2k}$, $M(k; L, N) = \binom{L}{k}$. However, this already fails at $N = 6$.

5 Strategy

We summarize the strategy of the proof. For $N, L \in \mathbb{Z}^+$, $L \geq 2N$, set

$$f(\lambda_1, \ldots, \lambda_N) = (\lambda_1 + i/2)^L \prod_{j \neq 1} (\lambda_1 - \lambda_j + i) - (\lambda_1 - i/2)^L \prod_{j \neq 1} (\lambda_1 - \lambda_j - i).$$

The polynomial $f(\lambda_1, \ldots, \lambda_N)$ is symmetric in the last $N-1$ variables, and the Bethe equations can be written as

$$f(\lambda_1, \ldots, \lambda_N) = 0, f(\lambda_2, \lambda_1, \ldots, \lambda_N) = 0, \ldots f(\lambda_N, \lambda_1, \ldots, \lambda_{N-1}) = 0.$$

For each root $\boldsymbol{\lambda}$ we can define a non-physical multiplicity such that the dimension of the Bethe vector at this root equals its multiplicity in the Bethe equations minus this non-physical multiplicity.

References

[1] W. Hao, R. I. Nepomechie, and A. J. Sommese, *Completeness of solutions of Bethe's equations*, Phys. Rev. E **88**, 052113 (2013); arXiv:1308.4645.

[2] N. A. Slavnov, *Algebraic Bethe ansatz*, 2018; https://api.semanticscholar.org/CorpusID:119586516.